Benchmarking Loss Functions for 12-Lead ECG Reconstruction from Limited Leads

A Multi-Objective Framework for Morphological Fidelity

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Point-wise accuracy is necessary but not sufficient: MSE anchors amplitude, correlation improves morphology *when calibrated*, and the best combination is architecture-dependent.

48	2,198	17	3
primary factorial models	held-out PTB-XL ECGs	paired stress conditions	external/device cohorts

Introduction & Objective

Reconstructing a clinical 12-lead ECG from leads **I, II, and V2** is underdetermined. A model can achieve high global correlation while distorting amplitude, QRS slopes, ST morphology, or downstream classifier outputs.

Question. Across U-Net, MultiScale-VAE, and ECG-AIM, what does each loss component contribute to waveform fidelity, morphology, robustness, and diagnostic preservation?

Core hypothesis

The losses are complementary constraints, not interchangeable surrogates. Their effects should therefore be estimated factorially and separately for each architecture.

Loss Anatomy: Complete 2^4 Objective

For target ECG y, reconstruction $\hat{y}$, L leads and T time samples, each binary mask $E/C/M/D$ activates one term:

$$\boxed{\mathcal{L} = E\,\mathcal{L}_{\text{MSE}} 0.5C\,\mathcal{L}_{\text{corr}} 0.1M\,\mathcal{L}_{\text{MMD}} 0.05D\,\mathcal{L}_{\Delta}}$$

$$\mathcal{L}_{\text{MSE}} = \frac{1}{LT}\|y-\hat{y}\|_2^2$$ amplitude calibration

$$\mathcal{L}_{\text{corr}} = 1 - \frac{1}{L}\sum_{\ell=1}^{L}\rho(y_\ell,\hat{y}_\ell)$$ shape / phase alignment

$$\mathcal{L}_{\Delta} = \text{MSE}(\Delta y, \Delta\hat{y})$$ slope preservation

$$\mathcal{L}_{\text{MMD}} = \mathbb{E}k(\hat{y},\hat{y}') + \mathbb{E}k(y,y') - 2\mathbb{E}k(\hat{y},y')$$ distribution matching

MMD uses four scale-adaptive RBF kernels with bandwidth multipliers $\{0.5, 1, 2, 4\}$ around the detached median batch distance. Distances are mean-squared per feature, preventing kernel underflow at ECG dimensionality.

E	C	M	D	Mask example
1	0	0	0	MSE-only 1000
1	1	1	1	full 1111
0	1	1	1	no-MSE 0111

Experimental Design

Factorial grid	3 architectures $\times$ 16 E/C/M/D masks $= 48$ seed-42 runs
Confirmation	18 runs: base, full, and validation-selected configurations at seeds 1337 and 2026
Input / target	I, II, V2 observed; nine missing leads evaluated primarily
Checkpoint rule	Lowest missing-lead validation MSE; Pearson tie-breaker
Clean test	PTB-XL fold 10: 2,198 ECGs from 1,904 patients
Statistics	2,000 paired patient-cluster BCa resamples; $\alpha = 0.0167$ for QRS, ST, diagnostic utility
Stress test	Gaussian + NSTDB BW/EM/MA at 24/12/6/0 dB, plus synthetic baseline drift
External/device	EchoNext $n = 5{,}442$; smartwatch $n = 719$; Sunnybrook Cath Lab $n = 20$
Frozen readouts	ECGFounder 150-task, five PTB-XL superclasses, and signal-quality head

Observed leads are preserved exactly by the unified adapter. Identical ECG IDs and deterministic noise realizations are used for paired comparisons.

Evaluation Contract

- **Signal:** MSE, RMSE, MAE, R^2, Pearson, SNR, derivative error.
- **Morphology:** standardized QRS $[-50, +100]$ ms; ST $[+120, +240]$ ms; J-point and R-peak timing.
- **Utility:** AUROC, AP, F1, ECE, Brier, subgroup gaps.
- **Transfer:** physical units and source sampling preserved; no per-record min–max transform.

Model Families & Reproducibility

Family	Batch	Epochs	Role in design
U-Net	256	10	deterministic convolutional baseline
MultiScale-VAE	32	10	variational, multi-resolution decoder
ECG-AIM	32	10	learned-token latent reconstruction

- ✓ 48/48 primary checkpoints load through one adapter
- ✓ finite $[B, 12, 5000]$ output and exact observed-lead passthrough
- ✓ 2,198 clean records and 17 stress conditions per cell
- ✓ strict JSON, per-record Parquet, source hashes, and 22/22 package checks

Results I: the 48-Cell Loss Landscape

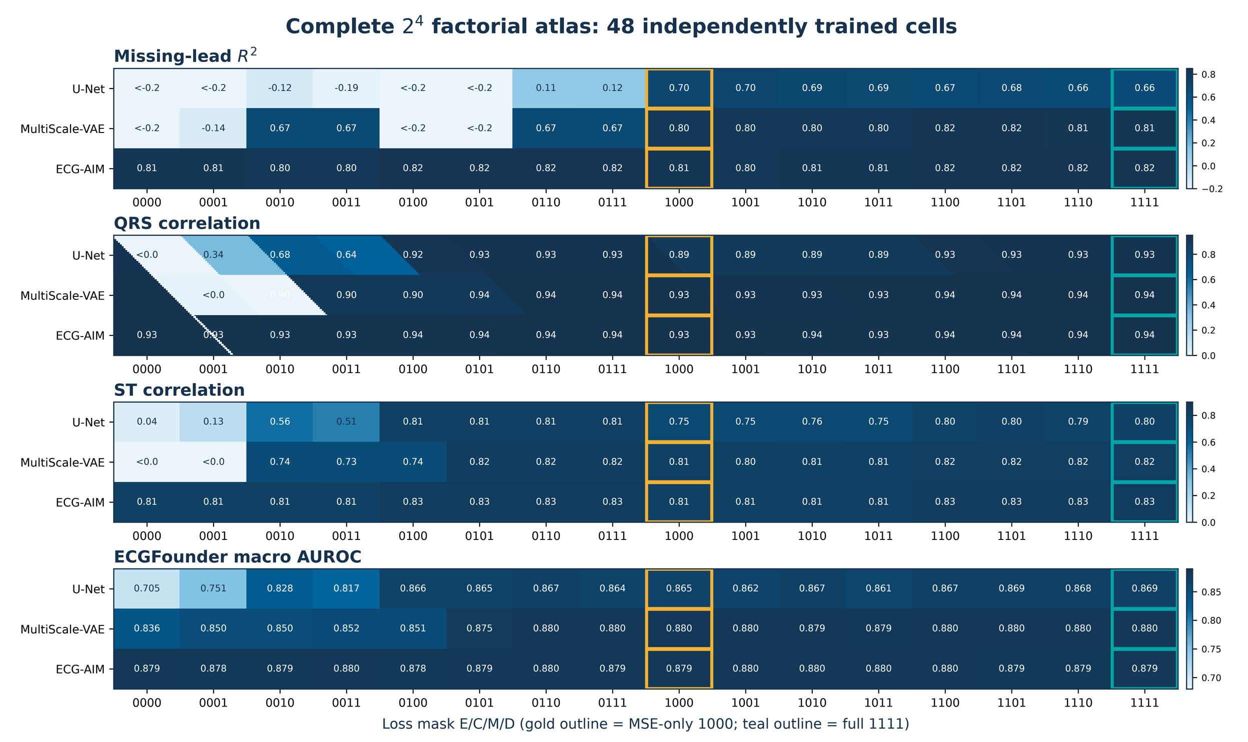

Figure 1. Complete E/C/M/D grid. The MSE-off half exposes severe calibration failures in U-Net and MultiScale-VAE; ECG-AIM is substantially less sensitive. Gold = prespecified MSE-only; teal = full composite.

Results II: Clinical Endpoint Tests

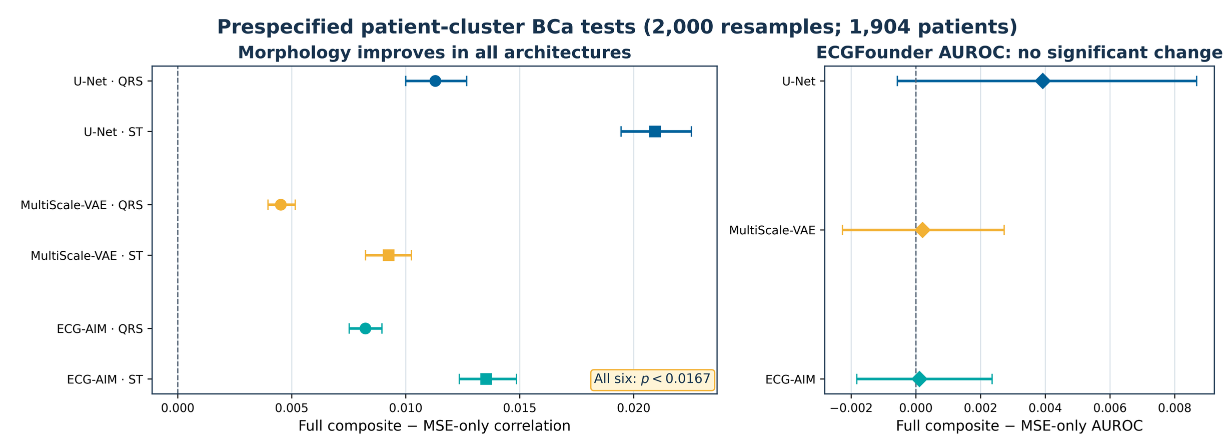

Figure 2. Full composite versus MSE-only. All six QRS/ST contrasts improve at the family-wise threshold. ECGFounder AUROC intervals cross zero in all architectures: no significant change, *not* proof of equivalence.

Critical failure mode. MultiScale-VAE mask 0100 (correlation only) attains Pearson $\rho = 0.874$ but $R^2 = -14.984$. Correlation can recover shape while allowing catastrophic scale error; **MSE is the calibration anchor**.

Architecture-Specific Best Cells

Family	Best R^2 mask	R^2	Full QRS	Full ST
U-Net	1000	0.700	0.929	0.798
MultiScale-VAE	1101	0.816	0.940	0.822
ECG-AIM	0111	**0.822**	**0.942**	**0.832**

There is **no universal winning mask**. U-Net favors MSE-only for R^2; MultiScale-VAE favors MSE+correlation+derivative; ECG-AIM's best R^2 is achieved without MSE. Endpoint choice must be declared before selecting a loss.

Representative External Reconstructions

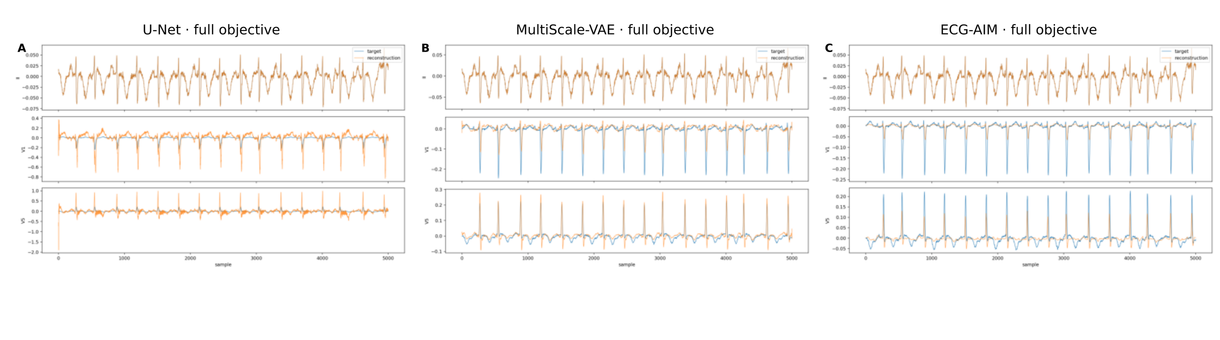

Figure 3. EchoNext targets (blue) and full-loss reconstructions (orange). Trace inspection reveals architecture-specific amplitude and morphology errors that aggregate metrics alone can hide.

Conditional Component Effects

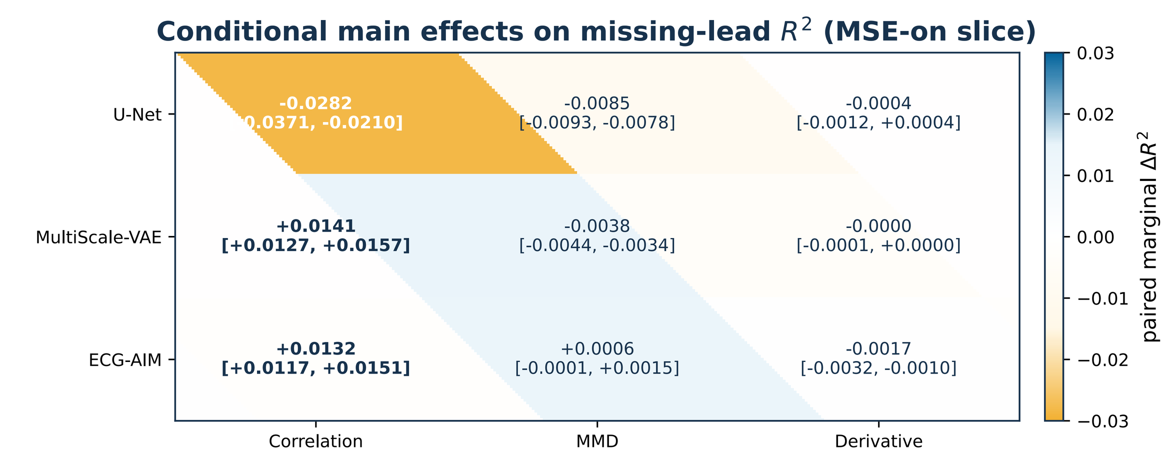

Figure 4. Paired marginal effects within the prespecified MSE-on 2^3 slice. Correlation lowers U-Net R^2 but raises MultiScale-VAE and ECG-AIM R^2; this sign reversal is direct evidence against a universal auxiliary-loss recipe. Values show estimate and 95% BCa interval.

Results III: Noise Robustness

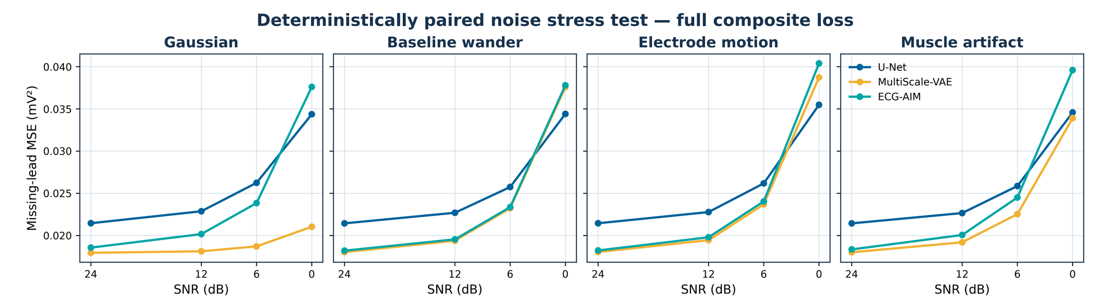

Figure 5. Identical paired noise realizations for each model. MultiScale-VAE has the lowest full-loss missing-lead MSE across the plotted stress curves, while all families degrade monotonically as SNR falls.

Results IV: External Transfer

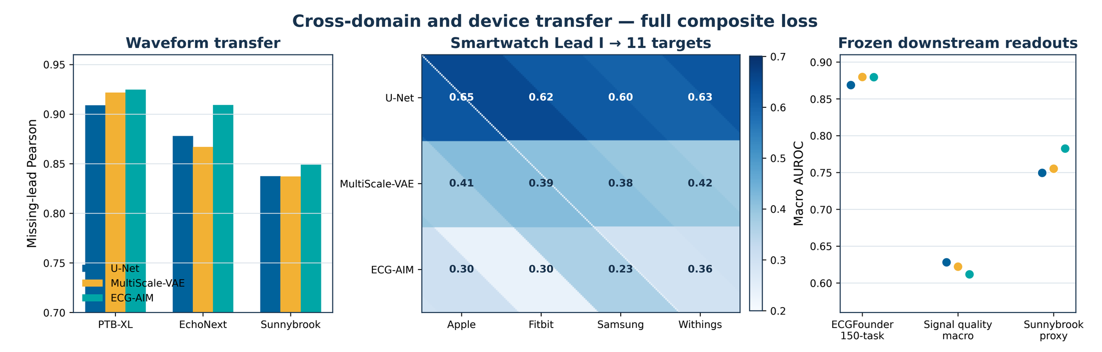

Figure 6. Full-loss models across waveform, device, and frozen downstream endpoints. EchoNext and Sunnybrook reveal smaller but still strong waveform transfer; single-lead smartwatch transfer is strongly architecture-dependent.

Scope matters. Smartwatch signals are calibrated simulator/device recordings, not a human pathology cohort. Sunnybrook superclass AUROC uses non-adjudicated Philips statement proxies; ECGFounder there is probability fidelity only.

Diagnostic & Signal-Quality

Full-loss endpoint	U-Net	MS-VAE	ECG-AIM
ECGFounder macro AUROC	0.869	**0.880**	0.879
PTB-XL quality macro AUROC	**0.628**	0.622	0.612
Any-artifact AUROC	0.683	0.683	**0.697**
Quality probability fidelity	0.806	0.893	**0.934**
Sunnybrook Pearson	0.838	0.839	**0.849**

Across all 48 cells, 27 meet the exploratory ECGFounder noninferiority margin of 0.02: ECG-AIM 16/16, MultiScale-VAE 11/16, U-Net 0/16. Signal-quality electrode-problem AUROC remains exploratory because fold 10 has only three positives.

Discussion

1. **MSE governs identifiability of scale.** Shape-only objectives can look convincing under Pearson correlation while being physically wrong.
2. **Correlation supplies the strongest morphology signal** on the MSE-on slice, but its R^2 effect changes sign by architecture.
3. **Auxiliary terms interact.** Marginal effects cannot be transferred blindly between a convolutional U-Net, a variational multi-scale model, and ECG-AIM.
4. **Utility is comparatively stable.** Morphology improves with the full objective without a significant ECGFounder AUROC shift, but this is not a clinical-safety claim.

Conclusion

Benchmark losses as experimental factors—not as a single leaderboard. For limited-lead ECG reconstruction, a calibrated point-wise anchor plus explicit morphology constraints is the safest default, followed by architecture- and endpoint-specific validation.

Limitations, Declaration & Access

Limitations. Confirmation seeds cover only the MSE-on base/full/selected slice. EchoNext diagnostic outputs retain tabular metadata. Smartwatch inputs are simulator/device signals. Sunnybrook labels are device-statement proxies. This benchmark evaluates reconstruction fidelity and frozen readouts, not clinical deployment safety.

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Declaration of interest. Conflict-of-interest statement: author confirmation required before printing.

Data and provenance: PTB-XL, NSTDB, EchoNext, the PhysioNet ECG-capable smartwatch database, and measured Sunnybrook Cath Lab ECGs. Exact machine-readable results, per-record tables, scripts, and hashes are included in the project repository.

github.com/whiteblaze143/benchmarking_loss_functions_ecg_reconstruction